

The diagram illustrates the sulfur cycle, showing the interconversions of various sulfur species and the enzymes involved. The cycle is organized around a central hub of sulfur species, with arrows indicating the direction of conversion and the associated enzymes.

Central Sulfur Species:

- HS^- (top center)
- $\text{S}(0)$ (top right)
- SO_3^{2-} (middle left)
- $\text{S}_2\text{O}_3^{2-}$ (middle center)
- $\text{S}_4\text{O}_6^{2-}$ (middle right)
- SO_3^{2-} (bottom right)
- SO_4^{2-} (bottom center)
- APS (bottom left)

Enzymes and Pathways:

- Reduction of HS^- to $\text{S}(0)$:** FccAB/Sqr
- Oxidation of $\text{S}(0)$ to HS^- :** PsrABC/ShyABCD, SudAB/Npsr/Nsr
- Reduction of $\text{S}(0)$ to SO_3^{2-} :** SoxY, SoxB, SoxZ, SoxX, SoxAB, SoxCD, SoxEF, SoxGH, SoxIJ, SoxKL, SoxMN, SoxOP, SoxPQ, SoxR, SoxS, SoxT, SoxU, SoxV, SoxW, SoxX, SoxY, SoxZ, SoxAA, SoxAB, SoxAC, SoxAD, SoxAE, SoxAF, SoxAG, SoxAH, SoxAI, SoxAJ, SoxAK, SoxAL, SoxAM, SoxAN, SoxAO, SoxAP, SoxAQ, SoxAR, SoxAS, SoxAT, SoxAU, SoxAV, SoxAW, SoxAX, SoxAY, SoxAZ, SoxBA, SoxBB, SoxBC, SoxBD, SoxBE, SoxBF, SoxBG, SoxBH, SoxBI, SoxBJ, SoxBK, SoxBL, SoxBM, SoxBN, SoxBO, SoxBP, SoxBQ, SoxBR, SoxBS, SoxBT, SoxBU, SoxBV, SoxBW, SoxBX, SoxBY, SoxBZ, SoxCA, SoxCB, SoxCC, SoxCD, SoxCE, SoxCF, SoxCG, SoxCH, SoxCI, SoxCJ, SoxCK, SoxCL, SoxCM, SoxCN, SoxCO, SoxCP, SoxCQ, SoxCR, SoxCS, SoxCT, SoxCU, SoxCV, SoxCW, SoxCX, SoxCY, SoxCZ, SoxDA, SoxDB, SoxDC, SoxDD, SoxDE, SoxDF, SoxDG, SoxDH, SoxDI, SoxDJ, SoxDK, SoxDL, SoxDM, SoxDN, SoxDO, SoxDP, SoxDQ, SoxDR, SoxDS, SoxDT, SoxDU, SoxDV, SoxDW, SoxDX, SoxDY, SoxDZ, SoxEA, SoxEB, SoxEC, SoxED, SoxEE, SoxEF, SoxEG, SoxEH, SoxEI, SoxEJ, SoxEK, SoxEL, SoxEM, SoxEN, SoxEO, SoxEP, SoxEQ, SoxER, SoxES, SoxET, SoxEU, SoxEV, SoxEW, SoxEX, SoxEY, SoxEZ, SoxFA, SoxFB, SoxFC, SoxFD, SoxFE, SoxFF, SoxFG, SoxFH, SoxFI, SoxFJ, SoxFK, SoxFL, SoxFM, SoxFN, SoxFO, SoxFP, SoxFQ, SoxFR, SoxFS, SoxFT, SoxFU, SoxFV, SoxFW, SoxFX, SoxFY, SoxFZ, SoxGA, SoxGB, SoxGC, SoxGD, SoxGE, SoxGF, SoxGG, SoxGH, SoxGI, SoxGJ, SoxGK, SoxGL, SoxGM, SoxGN, SoxGO, SoxGP, SoxGQ, SoxGR, SoxGS, SoxGT, SoxGU, SoxGV, SoxGW, SoxGX, SoxGY, SoxGZ, SoxHA, SoxHB, SoxHC, SoxHD, SoxHE, SoxHF, SoxHG, SoxHH, SoxHI, SoxHJ, SoxHK, SoxHL, SoxHM, SoxHN, SoxHO, SoxHP, SoxHQ, SoxHR, SoxHS, SoxHT, SoxHU, SoxHV, SoxHW, SoxHX, SoxHY, SoxHZ, SoxIA, SoxIB, SoxIC, SoxID, SoxIE, SoxIF, SoxIG, SoxIH, SoxII, SoxIJ, SoxIK, SoxIL, SoxIM, SoxIN, SoxIO, SoxIP, SoxIQ, SoxIR, SoxIS, SoxIT, SoxIU, SoxIV, SoxIW, SoxIX, SoxIY, SoxIZ, SoxJA, SoxJB, SoxJC, SoxJD, SoxJE, SoxJF, SoxJG, SoxJH, SoxJI, SoxJJ, SoxJK, SoxJL, SoxJM, SoxJN, SoxJO, SoxJP, SoxJQ, SoxJR, SoxJS, SoxJT, SoxJU, SoxJV, SoxJW, SoxJX, SoxJY, SoxJZ, SoxKA, SoxKB, SoxKC, SoxKD, SoxKE, SoxKF, SoxKG, SoxKH, SoxKI, SoxKJ, SoxKK, SoxKL, SoxKM, SoxKN, SoxKO, SoxKP, SoxKQ, SoxKR, SoxKS, SoxKT, SoxKU, SoxKV, SoxKW, SoxKX, SoxKY, SoxKZ, SoxLA, SoxLB, SoxLC, SoxLD, SoxLE, SoxLF, SoxLG, SoxLH, SoxLI, SoxLJ, SoxLK, SoxLL, SoxLM, SoxLN, SoxLO, SoxLP, SoxLQ, SoxLR, SoxLS, SoxLT, SoxLU, SoxLV, SoxLW, SoxLX, SoxLY, SoxLZ, SoxMA, SoxMB, SoxMC, SoxMD, SoxME, SoxMF, SoxMG, SoxMH, SoxMI, SoxMJ, SoxMK, SoxML, SoxMN, SoxMO, SoxMP, SoxMQ, SoxMR, SoxMS, SoxMT, SoxMU, SoxMV, SoxMW, SoxMX, SoxMY, SoxMZ, SoxNA, SoxNB, SoxNC, SoxND, SoxNE, SoxNF, SoxNG, SoxNH, SoxNI, SoxNJ, SoxNK, SoxNL, SoxNM, SoxNO, SoxNP, SoxNQ, SoxNR, SoxNS, SoxNT, SoxNU, SoxNV, SoxNW, SoxNX, SoxNY, SoxNZ, SoxOA, SoxOB, SoxOC, SoxOD, SoxOE, SoxOF, SoxOG, SoxOH, SoxOI, SoxOJ, SoxOK, SoxOL, SoxOM, SoxON, SoxOO, SoxOP, SoxOQ, SoxOR, SoxOS, SoxOT, SoxOU, SoxOV, SoxOW, SoxOX, SoxOY, SoxOZ, SoxPA, SoxPB, SoxPC, SoxPD, SoxPE, SoxPF, SoxPG, SoxPH, SoxPI, SoxPJ, SoxPK, SoxPL, SoxPM, SoxPN, SoxPO, SoxPP, SoxPQ, SoxPR, SoxPS, SoxPT, SoxPU, SoxPV, SoxPW, SoxPX, SoxPY, SoxPZ, SoxQA, SoxQB, SoxQC, SoxQD, SoxQE, SoxQF, SoxQG, SoxQH, SoxQI, SoxQJ, SoxQK, SoxQL, SoxQM, SoxQN, SoxQO, SoxQP, SoxQQ, SoxQR, SoxQS, SoxQT, SoxQU, SoxQV, SoxQW, SoxQX, SoxQY, SoxQZ, SoxRA, SoxRB, SoxRC, SoxRD, SoxRE, SoxRF, SoxRG, SoxRH, SoxRI, SoxRJ, SoxRK, SoxRL, SoxRM, SoxRN, SoxRO, SoxRP, SoxRQ, SoxRR, SoxRS, SoxRT, SoxRU, SoxRV, SoxRW, SoxRX, SoxRY, SoxRZ, SoxSA, SoxSB, SoxSC, SoxSD, SoxSE, SoxSF, SoxSG, SoxSH, SoxSI, SoxSJ, SoxSK, SoxSL, SoxSM, SoxSN, SoxSO, SoxSP, SoxSQ, SoxSR, SoxSS, SoxST, SoxSU, SoxSV, SoxSW, SoxSX, SoxSY, SoxSZ, SoxTA, SoxTB, SoxTC, SoxTD, SoxTE, SoxTF, SoxTG, SoxTH, SoxTI, SoxTJ, SoxTK, SoxTL, SoxTM, SoxTN, SoxTO, SoxTP, SoxTQ, SoxTR, SoxTS, SoxTT, SoxTU, SoxTV, SoxTW, SoxTX, SoxTY, SoxTZ, SoxUA, SoxUB, SoxUC, SoxUD, SoxUE, SoxUF, SoxUG, SoxUH, SoxUI, SoxUJ, SoxUK, SoxUL, SoxUM, SoxUN, SoxUO, SoxUP, SoxUQ, SoxUR, SoxUS, SoxUT, SoxUU, SoxUV, SoxUW, SoxUX, SoxUY, SoxUZ, SoxVA, SoxVB, SoxVC, SoxVD, SoxVE, SoxVF, SoxVG, SoxVH, SoxVI, SoxVJ, SoxVK, SoxVL, SoxVM, SoxVN, SoxVO, SoxVP, SoxVQ, SoxVR, SoxVS, SoxVT, SoxVU, SoxVV, SoxVW, SoxVX, SoxVY, SoxVZ, SoxWA, SoxWB, SoxWC, SoxWD, SoxWE, SoxWF, SoxWG, SoxWH, SoxWI, SoxWJ, SoxWK, SoxWL, SoxWM, SoxWN, SoxWO, SoxWP, SoxWQ, SoxWR, SoxWS, SoxWT, SoxWU, SoxWV, SoxWW, SoxWX, SoxWY, SoxWZ, SoxXA, SoxXB, SoxXC, SoxXD, SoxXE, SoxXF, SoxXG, SoxXH, SoxXI, SoxXJ, SoxXK, SoxXL, SoxXM, SoxXN, SoxXO, SoxXP, SoxXQ, SoxXR, SoxXS, SoxXT, SoxXU, SoxXV, SoxXW, SoxXX, SoxXY, SoxXZ, SoxYA, SoxYB, SoxYC, SoxYD, SoxYE, SoxYF, SoxYG, SoxYH, SoxYI, SoxYJ, SoxYK, SoxYL, SoxYM, SoxYN, SoxYO, SoxYP, SoxYQ, SoxYR, SoxYS, SoxYT, SoxYU, SoxYV, SoxYW, SoxYX, SoxYY, SoxYZ, SoxZA, SoxZB, SoxZC, SoxZD, SoxZE, SoxZF, SoxZG, SoxZH, SoxZI, SoxZJ, SoxZK, SoxZL, SoxZM, SoxZN, SoxZO, SoxZP, SoxZQ, SoxZR, SoxZS, SoxZT, SoxZU, SoxZV, SoxZW, SoxZX, SoxZY, SoxZZ

Dissimilatory sulfate reduction pathway

The diagram illustrates the dissimilatory sulfate reduction pathway, showing the flow of electrons and the conversion of various intermediates across the membrane.

Cytoplasm:

- SO_4^{2-} is reduced to **APS** by the enzyme **Sat**.
- APS** is reduced to SO_3^{2-} by the enzyme **AprBA**.
- SO_3^{2-} enters the membrane.

Membrane:

- QmoABC** (green box) is coupled with the reduction of MQH_2 to **MQ** and provides electrons (e^-) to **AprBA**.
- DsrMKJOP** (green box) is coupled with the reduction of MQH_2 to **MQ** and provides electrons to **DsrC**.

Periplasm:

- DsrC** (blue circle) is a cytochrome c_1 that exists in two states: a reduced state with two thiol groups (SH) and an oxidized state with a disulfide bond (S-S) and a red **S(0)** atom.
- The cycle involves the reduction of SO_3^{2-} to HS^- by **DsrBA** (yellow box), followed by the oxidation of HS^- to S(0) by **DsrC**, and the subsequent oxidation of S(0) to $\text{S}_2\text{O}_3^{2-}$ by **DsrC**.

Legend:

- Cytoplasm** (top)
- Membrane** (middle)
- Periplasm** (bottom)

Dissimilatory sulfide oxidation pathway